

SVEditor: For All Your Verilog and SystemVerilog Development Needs

This article focuses on a software that can be used for Verilog and SystemVerilog development projects. As it has been included in this month's DVD of EFY Plus, let us take a look at its features by installing it and setting up a project with it

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SVEditor is an Eclipse-based IDE (integrated development environment) for SystemVerilog and Verilog development. It provides a colourising editor for SystemVerilog with support for source navigation, content assist, source indent and auto-indent, SystemVerilog source templates and context-sensitive viewing of source documentation. Users have praised SVEditor for some of its features, such as searching for a colourisation of SystemVerilog keywords or words in a directory, auto-completion and some content assist. They believe you can use SVEditor for debugging if you love emacs. Quoting from sourceforge.net, one user has said, "I'm totally stoked that this project is under active development. The tool is already very useful. Keep up the good work!"

Features

SVEditor provides a variety of features that make developing designs and test-bench environments in SystemVerilog simpler and more efficient. The software provides many ways to navigate across the project source. The 'Open Declaration' option navigates to the location of the item's declaration like class, module, interface, field or method. By click-

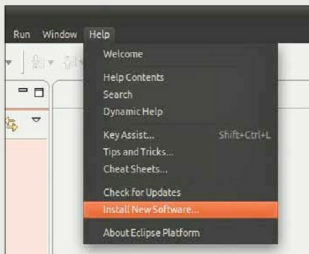


Fig. 1: Install New Software menu

Features at a glance

- SystemVerilog source index
- Syntax colouring editor
- Content assist
- Cross-linking between data structure usage and declaration
- SystemVerilog editor
- Class Hierarchy View

ing the 'Open Type Hierarchy' tab, it opens the 'Type Hierarchy' view for the specified class or module. Another feature, called 'Doc Hover', displays the code documentation when hovering around the documented element. The New Class/Interface/Module Wizards can be used to quickly create

new files containing a class, interface, or module. The 'Open Class Diagram' displays a UML-style class diagram of the target class.

How to install SVEditor

Listed below are several ways to install SVEditor.

A combination of Eclipse and SVEditor, SVE provides a complete development environment

for SystemVerilog. It is provided as an installable package for Debian and the Windows operating systems. If you are not already an Eclipse user, SVE is the simplest way to get started with SVEditor.

Installing SVEditor in an existing Eclipse environment. Via the SVEditor Update Site, SVEditor can be installed into an existing Eclipse environment. All you need to do is to open the 'Install New Software' dialogue box by selecting the 'Help-Install New Software' option (in Eclipse) as shown in the Fig. 1. As you can see in the figure, a new dialogue box opens up. Then, select 'Add' and specify the URL of the